DO NOW — RECALL

Lesson 084 · A* Search: g, h and f Scores

Name: _____ Class: _____ Date: _____

1. What is Dijkstra's tentative distance?

2. What does f(n) equal in A*?

3. What is an admissible heuristic?

4. Binary search target12 in sorted [2,4,6,8,10,12,14]. Show midpoint indexes.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Known best route cost from source.

2. Model answer

g(n)+h(n).

3. Model answer

Never overestimates true remaining cheapest cost.

4. Model answer

mid3=8 then mid5=12; return index5.

By the end, you can…

• A* orders candidate states using f(n)=g(n)+h(n), where g is known cost from the start and h estimates remaining cost to the goal.

• A heuristic that never overestimates remaining true cost is admissible; with suitable search implementation/assumptions this supports optimality.

• Track open candidates, their g/h/f and predecessors. An overestimating heuristic or incorrect update can make search miss the shortest path.

Vocabulary — know these terms

A* — Heuristic-guided shortest-path search.

g(n) — Cost so far from start.

h(n) — Estimated cost to goal.

f(n) — g(n)+h(n) priority.

NEW LEARNING

Read, annotate and explain

Observe

A* orders candidate states using f(n)=g(n)+h(n), where g is known cost from the start and h estimates remaining cost to the goal.

Apply

A heuristic that never overestimates remaining true cost is admissible; with suitable search implementation/assumptions this supports optimality.

Evaluate

Track open candidates, their g/h/f and predecessors. An overestimating heuristic or incorrect update can make search miss the shortest path.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Find P→R: P→Q = 3, P→R = 8, Q→R = 2. Heuristics: h(P)=5, h(Q)=2, h(R)=0.

Calculate all frontier f-scores

From P: Q has g=3,h=2,f=5 ; R has g=8,h=0,f=8 . Put both in OPEN. Choose Q next (smallest f), not simply the nearest-looking node.

Improve the path to R

Via Q→R, new g(R)=3+2=5 , which improves the existing 8. Thus f(R)=5+0=5 , and predecessor R becomes Q. OPEN now contains R with the improved values.

Confirm the path and heuristic meaning

Select goal R from OPEN, then reconstruct R←Q←P to obtain P→Q→R , cost 5 . The supplied heuristics do not overestimate remaining true cost; the next lesson can explore what happens when this fails.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

After expanding A, calculate f for B and C using stated graph/heuristics. [6]

Question 2

After expanding B then C, show improvements to g(C) and g(D); state best path. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare A* with Dijkstra when h=0 everywhere, and explain risk of overestimation. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A* h(n) is the exact distance to the goal for every possible heuristic.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.